AQI PM2.5 Demo

- [x] Show raw tables

Prediction rows 1000

- [x] Enable next-hour forecast
- [ ] Use full predictions (slow)

AQI Big Data (PM2.5) Dashboard

Data source: Hopsworks Feature Store | Model: linear_regression

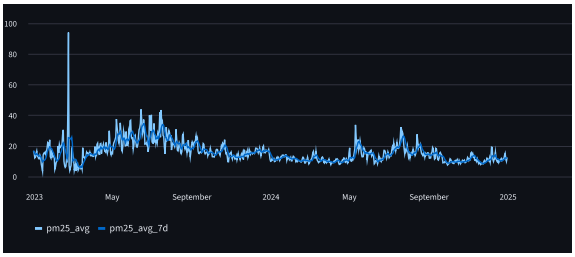
Model Metrics

	model	rmse	mae	r2
0	linear_regression	6.8967	1.9807	0.9064

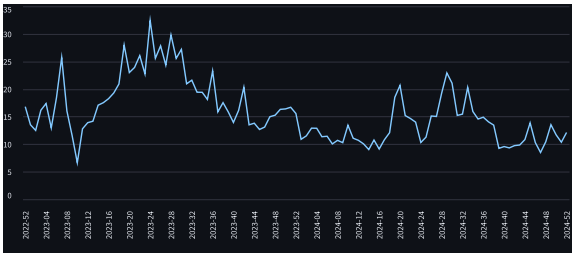
Best Model: linear_regression | R²: 0.9064 | RMSE: 6.90

EDA Trends

Daily Trend (with 7-day missing average)



Weekly Trend

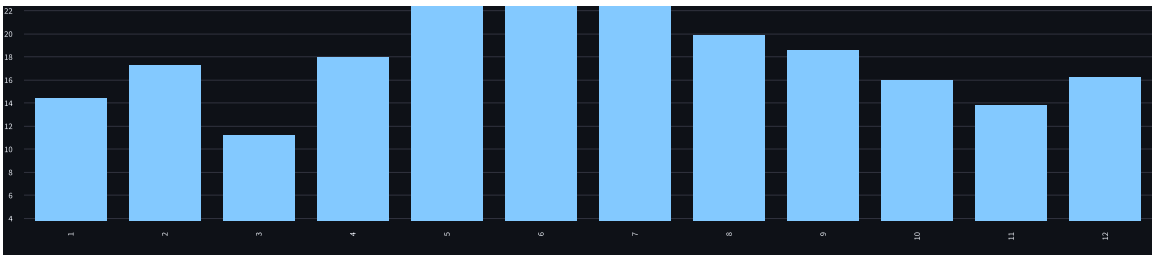


	date	pm25_avg	n	pm25_avg_7d
0	2022-12-31 00:00:00	16.8099	276	16.8099
1	2023-01-01 00:00:00	15.5845	857	16.1972
2	2023-01-02 00:00:00	12.1845	826	14.8596
3	2023-01-03 00:00:00	12.7102	868	14.3223
4	2023-01-04 00:00:00	15.144	868	14.4866
5	2023-01-05 00:00:00	16.699	887	14.8554
6	2023-01-06 00:00:00	13.2181	867	14.6215
7	2023-01-07 00:00:00	12.8227	832	14.0519
8	2023-01-08 00:00:00	11.6641	829	13.4918
9	2023-01-09 00:00:00	11.5807	875	13.4055

	year	week_of_year	pm25_avg	n	year_week
20	2023	20	23.0034	5634	2023-20
21	2023	21	23.9278	5205	2023-21
22	2023	22	26.0697	5529	2023-22
23	2023	23	22.7531	5939	2023-23
24	2023	24	32.4219	5207	2023-24
25	2023	25	25.5935	4774	2023-25
26	2023	26	27.8394	4863	2023-26
27	2023	27	24.3477	5000	2023-27
28	2023	28	29.8163	3114	2023-28
29	2023	29	25.5726	5270	2023-29

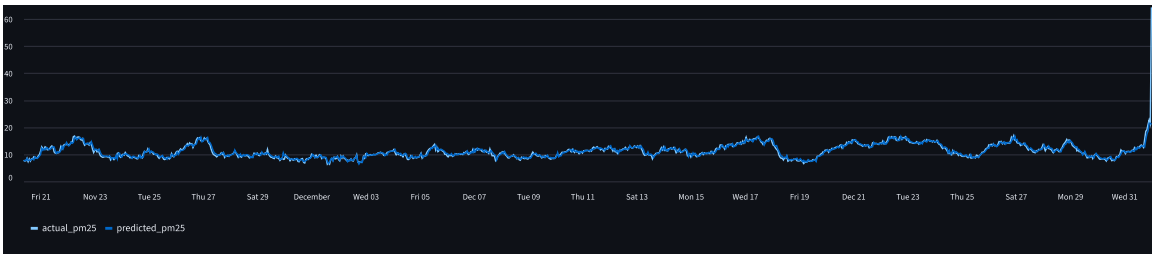
Yearly PM2.5 View

Year: 2023



Total days: 18.78 | Total obs: 94.00 | Total pm25: 3.60

Predictions (Actual vs Predicted)



	date	actual_pm25	predicted_pm25
36814	2025-11-20 09:00:00	7.8516	7.5727
36815	2025-11-20 10:00:00	7.558	7.7523
36816	2025-11-20 11:00:00	7.6826	7.7603
36817	2025-11-20 12:00:00	8.5953	8.476
36818	2025-11-20 13:00:00	7.3395	7.7091
36819	2025-11-20 14:00:00	8.3883	7.9289
36820	2025-11-20 15:00:00	7.5392	7.775
36821	2025-11-20 16:00:00	8.3953	8.1947
36822	2025-11-20 17:00:00	8.3301	8.3158
36823	2025-11-20 18:00:00	9.1433	7.9885

Next-hour Forecast (t+1)

Forecast (t+1)	Last Observed (t=0)	Next Hour (t+1)
468.101	508.402	2026-01-01 08:00:00

Forecast uses the trained multivariate LSTM model and the latest observed pollutant values.

Target: Pm_aqi_cn_1h_ahead (AQI 1 hour ahead)